PHYS-467 Assignment 2 Solution

Questions 1 and 2 See code provided
Question 3

From Bayes rule, we have

$$P(\vec{s}|Y,\rho,p_e) = \frac{P(Y|\vec{s},\rho,p_e)P(\vec{s}|\rho,p_e)}{P(Y|\rho,p_e)} = \frac{P(Y|\vec{s},\rho,p_e)P(\vec{s})}{P(Y|\rho,p_e)} \tag{1}$$

First, the distribution of $\vec{s}$ does not depend on ρ or p_e, and its components are *independent and identically distributed* with $P(s_i = 1) = P(s_i = -1) = 1/2$, thus

$$P(\vec{s}) = \prod_i P(s_i) = \prod_i \frac{1}{2}\left(\delta(s_i = 1) + \delta(s_i = -1)\right) = \frac{1}{2^n}\prod_i \delta(s_i = \pm 1) \tag{2}$$

Secondy, note that the $Y_{i,j}$ are also *independent* and for $i < j$

$$P(Y_{iy} = y|\vec{s},\rho,p_e) = \begin{cases} p_e \times \rho \text{ if } Y_{ij} = s_i s_j \\ p_e \times (1-\rho) \text{ if } Y_{ij} = -s_i s_j \\ (1-p_e) \text{ if } Y_{ij} = 0 \end{cases} \tag{3}$$

And $Y_{i,j} = Y_{j,i}$. Finally, $Y_{i,i} = 0$. Hence :

$$\begin{aligned} P(Y|\vec{s},p_e,\rho) &= \prod_{i<j} (1-p_e) \times \delta(Y_{ij} = 0) + p_e(\rho\delta(Y_{ij} = s_i s_j) + (1-\rho)\delta(Y_{ij} = -s_i s_j)) \\ &\times \prod_i \delta(Y_{ii} = 0) \prod_{i>j} \delta(Y_{ij} = Y_{ji}) \\ &= \frac{1}{Z(Y,p_e)} \prod_{i<j} \rho\delta(Y_{ij} = s_i s_j) + (1-\rho)\delta(Y_{ij} = -s_i s_j) = \frac{1}{Z(Y,p_e)} \rho^{\sum_{i<j}\delta(Y_{ij} = -s_i s_j)} (1-\rho)^{\sum_{i<j}\delta(Y_{ij} = s_i s_j)} \end{aligned}$$

Hence

$$P(\vec{s}|Y,\rho,p_e) = \frac{\prod_i \delta(s_i = \pm 1) \times \rho^{\sum_{i<j}\delta(Y_{ij} = -s_i s_j)}(1-\rho)^{\sum_{i<j}\delta(Y_{ij} = s_i s_j)}}{2^n \times P(Y|\rho,p_e) \times Z(Y,p_e)} \tag{4}$$

Note that the numerator is independent of p_e, so the denominator (which is a normalization constant) is also independent of p_e. Thus, we have

$$P(\vec{s}|Y,\rho) = P(\vec{s}|Y,\rho,p_e) \propto \prod_i \delta(s_i = \pm 1) \times \rho^{\sum_{i<j}\delta(Y_{ij} = -s_i s_j)}(1-\rho)^{\sum_{i<j}\delta(Y_{ij} = s_i s_j)} \tag{5}$$

Question 4 Note x the index flipped $s'_x = -s_x$ We have

$$\frac{P(\vec{s'}|Y,\rho)}{P(\vec{s}|Y,\rho)} = \rho^{\sum_{i<j}\delta(Y_{ij} = -s'_i s'_j) - \delta(Y_{ij} = -s_i s_j)} \times (1-\rho)^{\sum_{i<j}\delta(Y_{ij} = s'_i s'_j) - \delta(Y_{ij} = s_i s_j)} \tag{6}$$

Now, note that $s_i s_j = s'_i s'_j$ if $i, j \neq x$. So, the expression reduces to

$$\frac{P(\vec{s'}|Y,\rho)}{P(\vec{s}|Y,\rho)} = \rho^{\sum_{i\neq x}\delta(Y_{xi} = -s'_x s'_i) - \delta(Y_{xi} = -s_x s_i)} \times (1-\rho)^{\sum_{i\neq x}\delta(Y_{xi} = s'_x s'_i) - \delta(Y_{xi} = s_x s_i)} \tag{7}$$

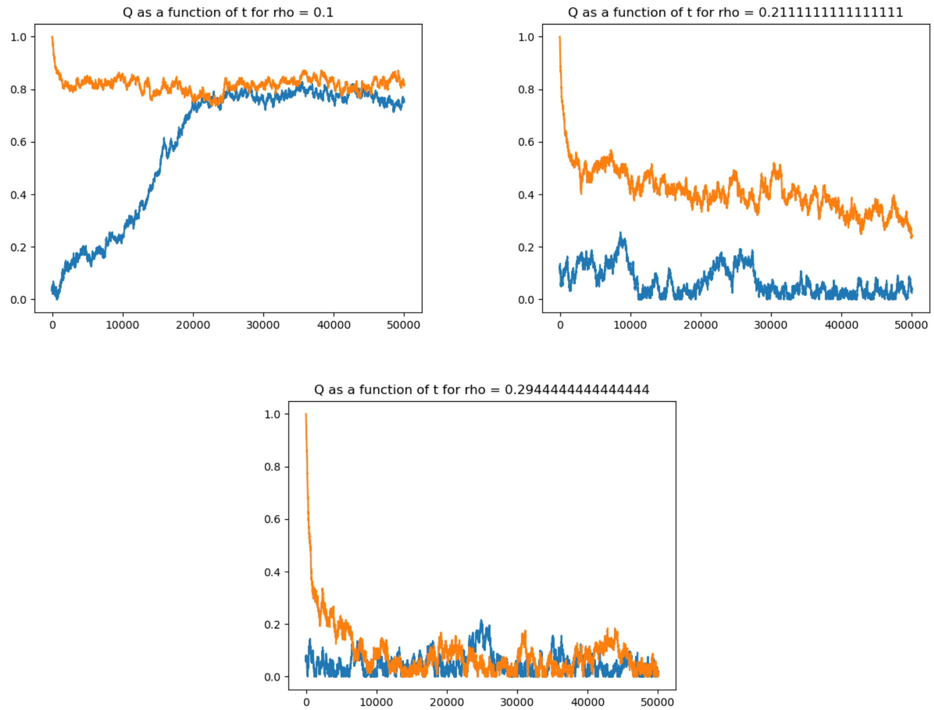

Figure 1: Overlap as a function of iteration for the planted (orange) and random (blue) initializations for $\rho = 0.1, 0.21, 0.29$. Note the divergence in thermalization time for $\rho = 0.21$

Note that for $Y_{ix} \neq 0$, $Y_{ix} = s_i s_x$ if and only if $Y_{ix} = -s'_i s'_x$. Thus, noting $n_+ = \sum_{i \neq x} \delta(Y_{ix} = s_i s_x)$ and $n_- = \sum_{i \neq x} \delta(Y_{ix} = -s_i s_x)$, we finally have

$$\frac{P(\vec{s'}|Y,\rho)}{P(\vec{s}|Y,\rho)} = \rho^{n_+ - n_-} \times (1-\rho)^{n_- - n_+} \tag{8}$$

See the code for the implementation.

Question 5

To identify the thermalization time, we run two Markov chains with 1) a random initialization and 2) a "planted" initialization where $s_0 = s$. We identify T_{therm} as the first time t such that the overlap $Q(s^t)$ is the same for both Markov chains (in practice, we will take the first time such that the overlaps differ by less than 0.01).

See Figure 2 for the plots of the thermalization time and overlap of s_{bo} as a function of ρ. We identify two phases : for small ρ, the thermalization time increases when ρ does. This is due to the fact that the recovery becomes more difficult as the graph becomes sparser. Then, at the transition (around $\rho \simeq 0.25$), T_{therm} seems to diverge. Finally, for larger ρ, the thermalization time decreases to values close to 0. On the other hand, we observe that after $\rho \simeq 0.25$, the overlap $Q(s_{BO})$ is close to 0. It then seems that when ρ is too large, the recovery of s becomes impossible, and the equilibrium distribution (which is uncorrelated with the ground truth) is reached quickly.

Question 6

Note that since we know that $s \in \{\pm 1\}^n$, s_{PCA} is defined as the *sign* of the leading eigenvector of Y. See Figure 3 for the performance of the two estimators. We observe on the plot that the Bayes optimal estimator and PCA behave very similarily. Due to the noisiness of the curves, it is not possible to conclude with certainty that BO performs better than PCA. This is surprising, as we would expect BO to perform much better. Moreover, note that PCA is much faster than MCMC, and thus might be preferable in practice. Finally, around $p_e = 4/n$, the overlap of both methods suddenly increases (while it stays very close to 0 below this p_e). This transition corresponds to the transition at $\rho = 0.25$ identified in the previous question.

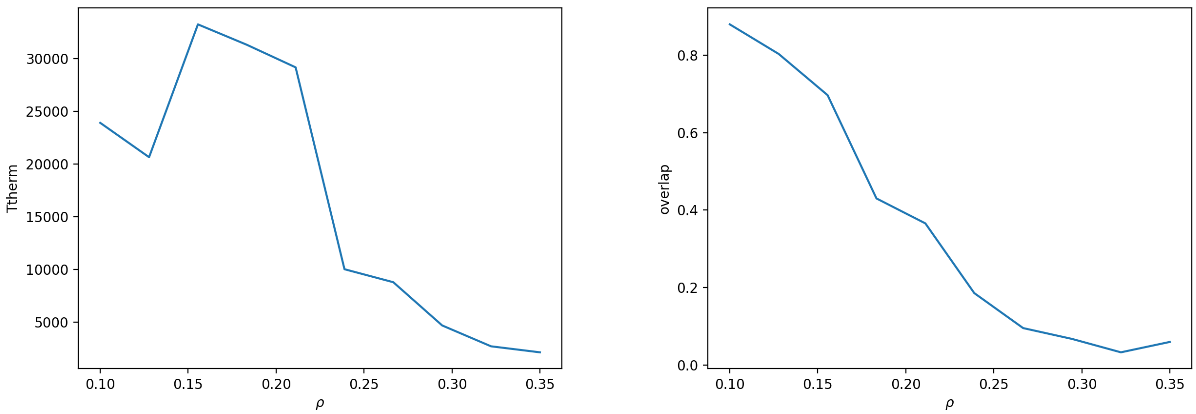

Figure 2: Thermalization time of MCMC (Left) and overlap of s_{BO} (Right) as a function of ρ with $n = 500$, $p_e = 4/n$. Averaged over 10 seeds.

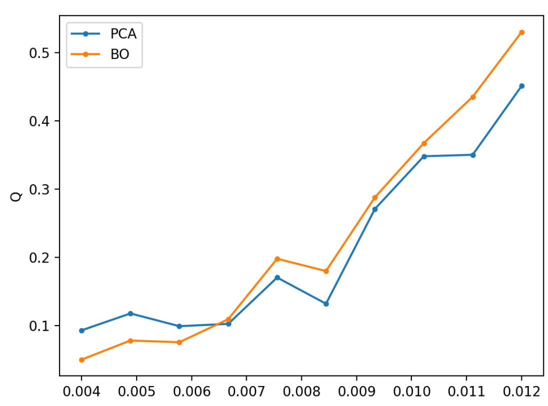

Figure 3: Overlap of s_{PCA} and s_{BO} as a function of p_e for fixed $\rho = 0.25$. Averaged over 20 random seeds.